Questions

Q1.

This question is about magnets and magnetism.

Figure 8 shows a magnet that has picked up three paper clips.

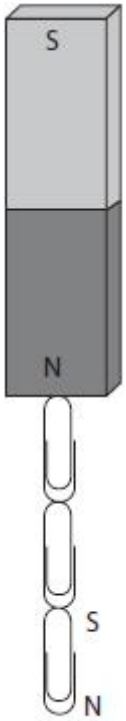


Figure 8

(i) The poles of the lowest paper clip are labelled.

Label the poles of the other two paper clips in Figure 8.

(2)

(ii) Complete the sentence, by choosing a word from the box, to describe the type of magnetism that these paper clips have.

(1)

alternated	earthed	induced	transformed

These paper clips have ..... magnetism.

(iii) Suggest a material that these paper clips in Figure 8 could be made from.

(1)

.....

(iv) When the paper clips were pulled off the magnet they fell separately to the table.

Describe how you could test whether any of the paper clips had kept any magnetism.

(2)

.....

.....

.....

.....

(Total for question = 6 marks)

Q2.

A student investigates the magnetic properties of three rods. Each rod is made of one of the following materials:

- soft iron
- steel
- wood

The student places each rod in a solenoid that is connected to a direct current power supply.

The power supply is switched on for a short time.

The student tests the magnetic strength of each rod by seeing how many paper clips it can pick up as shown in Figure 6.

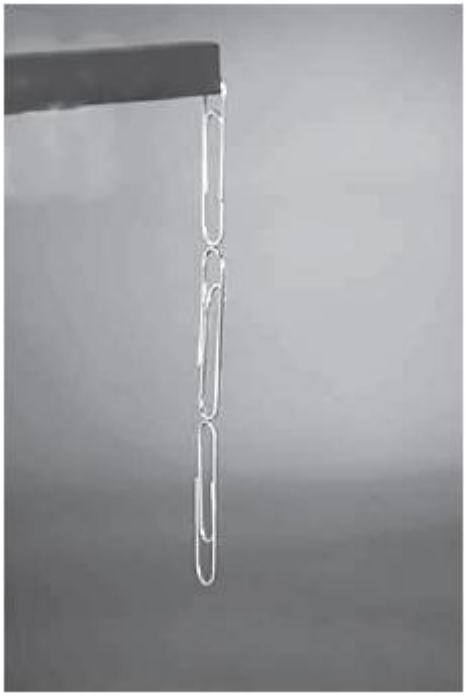


Figure 6

The student's results are shown in Figure 7.

	number of paper clips picked up by rod			
rod	before rod is placed in solenoid	when there is current in solenoid	1 minute after current is switched off	10 minutes after current is switched off
A	0	0	0	0
B	0	6	1	0
C	0	8	7	7

Figure 7

Complete the table below to show which material (soft iron, steel or wood) each rod is made from, with the reason why.

Part of the table has been done for you.

Use information from Figure 7.

(3)

rod	material	reason
A		It is not magnetic because it does not pick up paper clips whether there is a current or not.
B		
C		

(Total for question = 3 marks)

Q3.

Three of the following are magnetic materials.

Which of these is **NOT** a magnetic material?

(1)

- ☐ **A** cobalt
- ☐ **B** copper
- ☐ **C** iron
- ☐ **D** nickel

(Total for question = 1 mark)

Q4.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which of these is a magnetic material?

(1)

- ☐ **A** aluminium
- ☐ **B** plastic
- ☐ **C** steel
- ☐ **D** lead

(Total for question = 1 mark)

Q5.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these is a magnetic material?

(1)

- ☐ **A** aluminium
- ☐ **B** carbon
- ☐ **C** cobalt
- ☐ **D** copper

(Total for question = 1 mark)

Q6.

Figure 6 shows some objects and words describing these objects.

Draw one line from each object to its description.

(2)

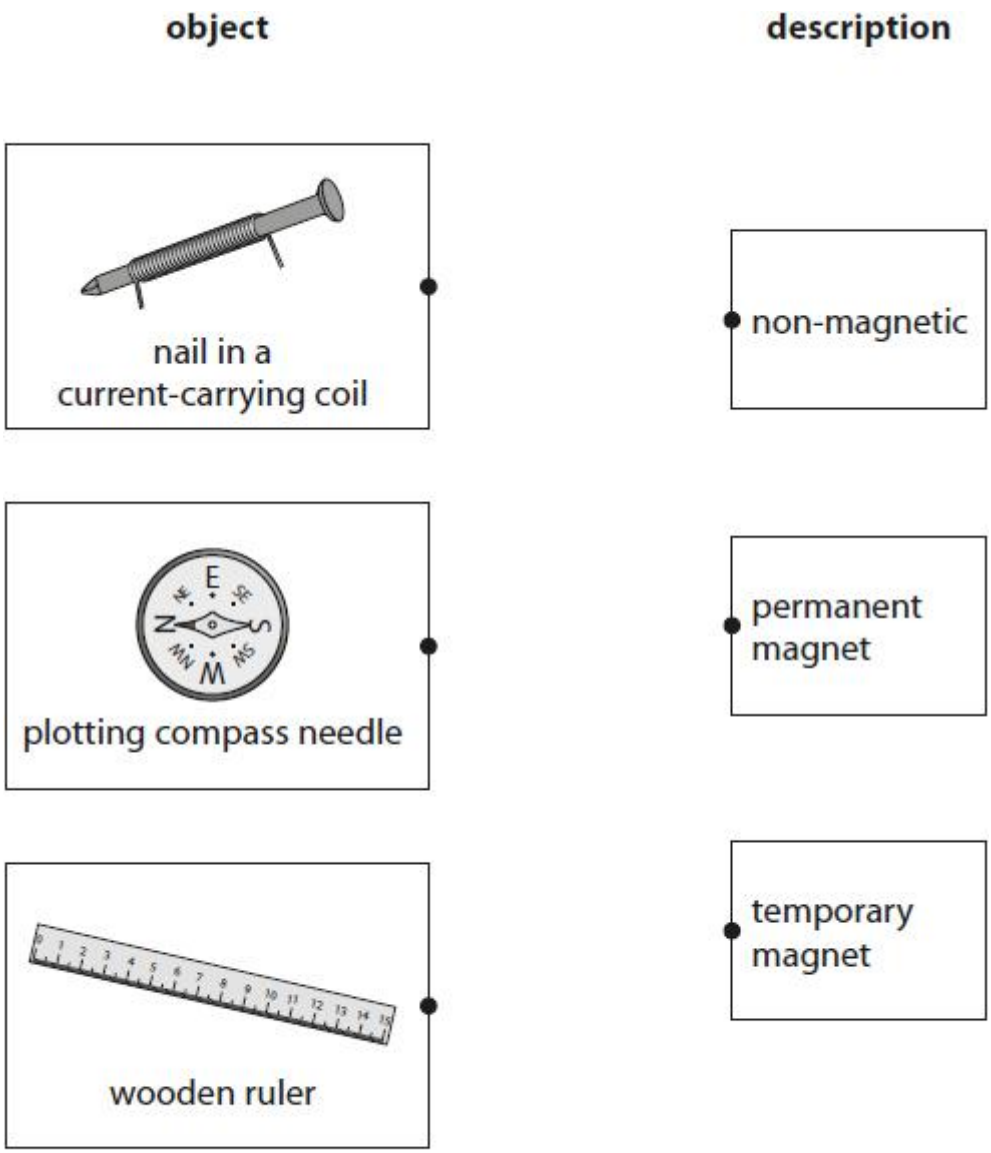


Figure 6

(Total for question = 2 marks)

Q7.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

A student is given two permanent magnets and some paper clips, as shown in Figure 10.

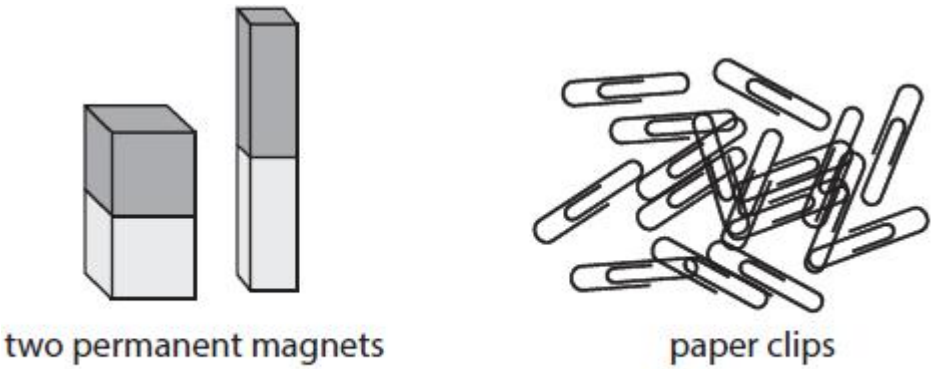


Figure 10

The paper clips are **not** magnets, but they are made from a magnetic material.

(i) Which of these is a magnetic material?

(1)

- ☒ **A** aluminium
- ☒ **B** iron
- ☒ **C** plastic
- ☒ **D** wood

(ii) Describe how the student could use the paper clips to find out which of the two permanent magnets is the stronger magnet.

(2)

.....

.....

.....

.....

.....

(Total for question = 3 marks)

Q8.

Figure 13 shows a part of a machine used to separate steel cans from aluminium cans.

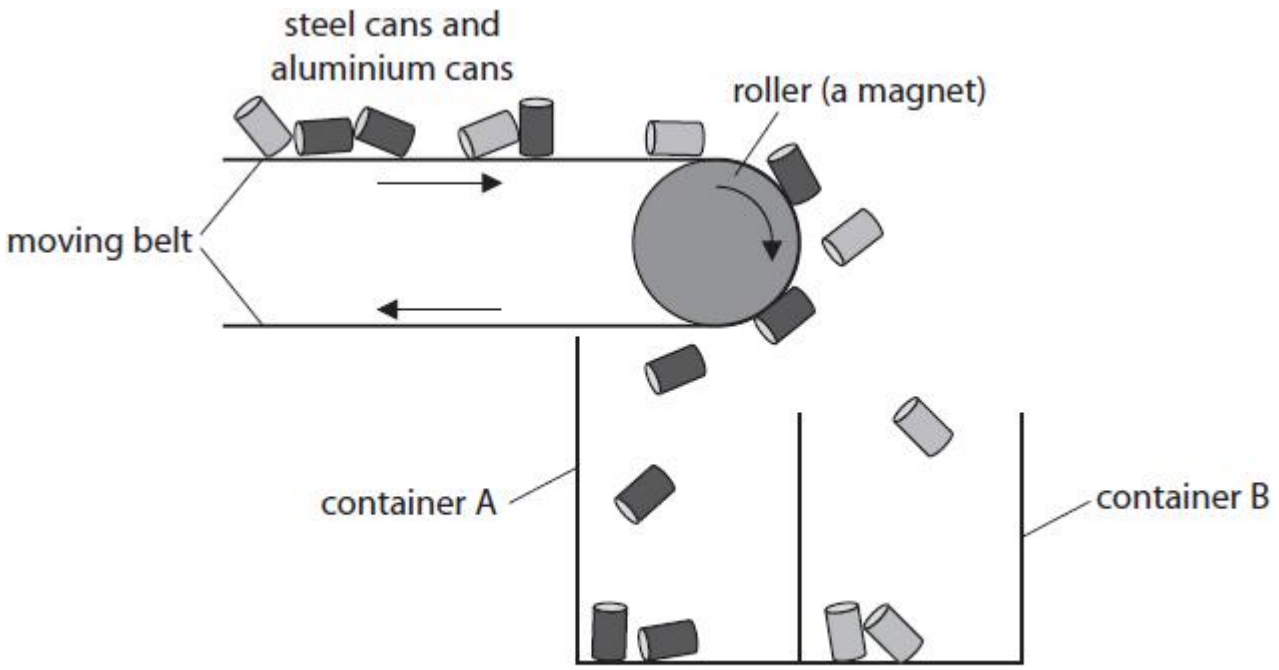


Figure 13

The cans are carried along a moving belt.

The belt goes around a roller.

The roller is a magnet.

Each can falls into one of the containers.

Explain how this machine separates the steel cans from the aluminium cans.

(2)

.....

.....

.....

.....

(Total for question = 2 marks)

Q9.

Figure 5 shows two magnetic poles facing each other.

The magnetic field between the poles is uniform.

On Figure 5, draw the magnetic field lines between the two poles and show the direction of this magnetic field.

(3)

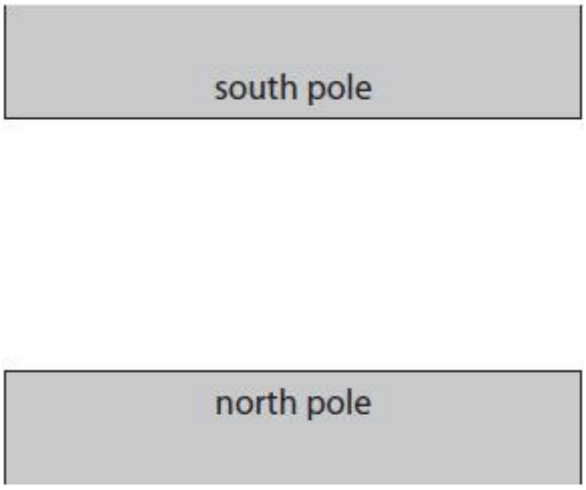


Figure 5

(Total for question = 3 marks)

Q10.

Figure 10 shows two magnetic poles facing each other.

There is a uniform magnetic field between the poles.

Draw field lines, on Figure 10, to show the uniform part of the magnetic field.
At least one line must have an arrow on it to indicate the direction of the magnetic field.

(2)

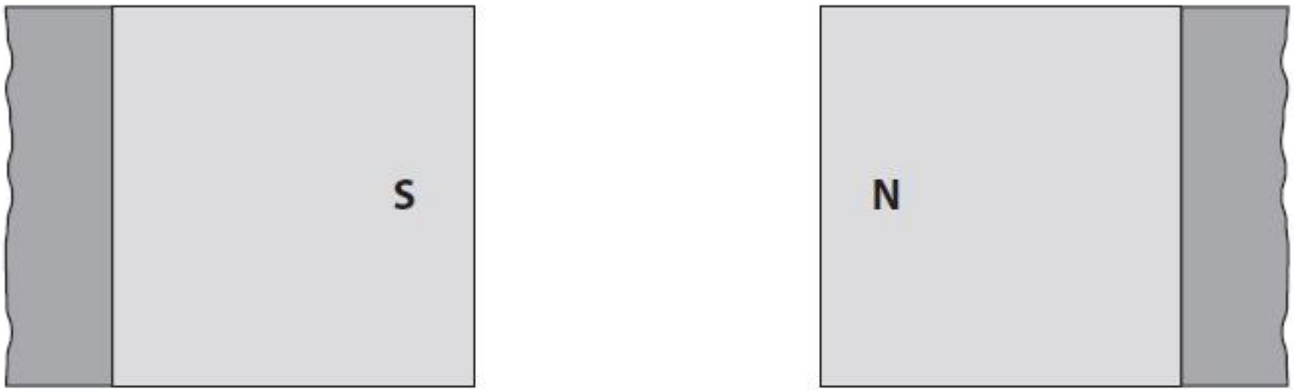
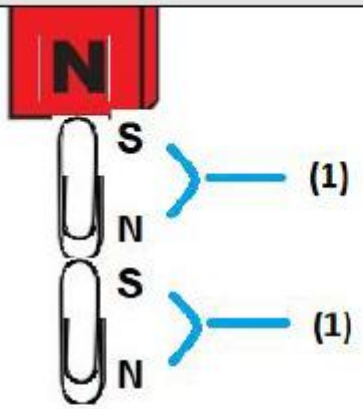


Figure 10

(Total for question = 2 marks)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
(i)		both poles needed for each mark (either side of paper clip, right or left) allow just S at the top of the pair and N at the bottom of the pair for 1 mark ignore the third paper clip after these two (given in question)	(2) AO1

Question number	Answer	Additional guidance	Mark
(ii)	induced (1)		(1) AO1

Question number	Answer	Additional guidance	Mark
(iii)	iron / steel / nickel / cobalt (1)	ignore 'metal' do not allow any other named metal	(1) AO1

Question number	Answer	Additional guidance	Mark
(iv)	description including **two** from use a (plotting) compass (1) (plotting compass) shows a change in direction / needle moves OR bring the paper clips together (1) seeing if they attract / repel (1)	 sees repulsion / repelling bring the paper clips near to a magnetic material ignore 'magnet' for this marking point do not accept 'attracts to a magnet'	**(2)** **AO1**
	OR use of iron filings (around the paperclips) (1) see a pattern (1)	 accept for two marks bring a magnet close to a paper clip to test for repulsion	

Q2.

Question number	Answer	Mark
	All three materials correctly identified (1) (rod A – wood, rod B – soft iron, rod C – steel) One mark for each reason: • rod B only attracts paper clips when there is a current in the coil (1) • rod C attracts paper clips when there is a current in the coil and for some time after (1)	**(3)**

Q3.

Question number	Answer	Mark
	B copper	**(1)**

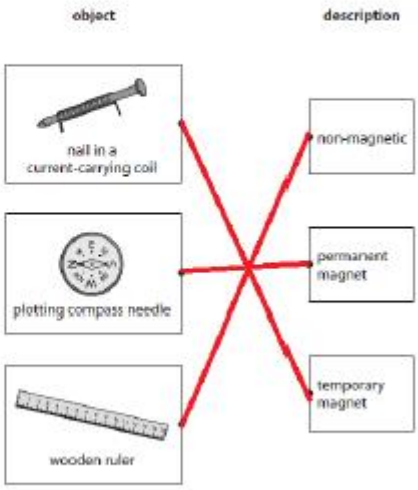
Q4.

Question Number	Answer	Mark
	C steel **A, B** and **D** are incorrect because they are non-magnetic	**(1)** **AO1.1**

Q5.

Question Number	Answer	Mark
	C cobalt **C** is the only correct answer. **A** is incorrect because aluminium is not magnetic. **B** is incorrect because carbon is not magnetic. **D** is incorrect because copper is not magnetic.	**(1)**

Q6.

Question number	Answer	Additional guidance	Mark
		three links correct (2) one link correct (1)	(2) AO1

Q7.

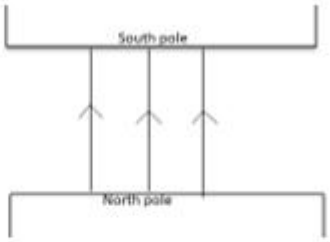
Question	Answer	Mark
(i)	**B** iron ***A,C and D*** *are not magnetic materials*	(1) AO1.1

Question	Answer	Additional guidance	Mark
(ii)	A description to include: method (1) put magnet near paper clips outcome (1) (picks up/attracts) more paper clips OR (has the) longest chain OR (picks up/attracts from a) greater distance	method can be included with the outcome in one sentence for two marks e.g. see which magnet attracts the most paper clips	**(2)** **AO1.2**

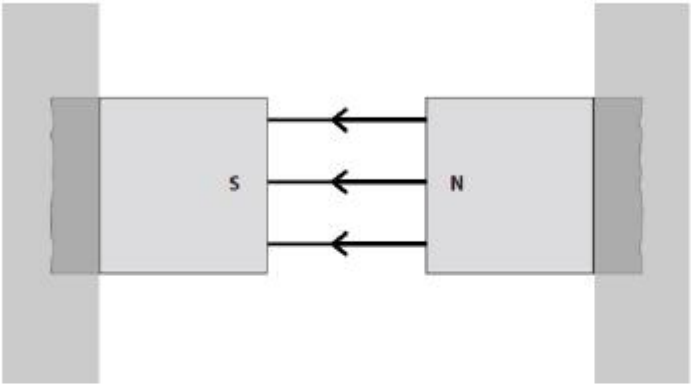
Q8.

Question number	Answer	Additional guidance	Mark
	an explanation linking any **two** of		**(2)**
	steel is magnetic (material) (1)	steel attracted / sticks to / carried round by magnet/roller)	
	aluminium is non-magnetic (material) (1)	is not attracted / does not stick (to magnet roller)	
	steel falls into container A / aluminium falls into container B (1)	steel cans are carried further round than aluminium and fall into A steel hangs on for longer / aluminium falls quicker	

Q9.

Question Number	Answer	Additional guidance	Mark
	 MP1: any (vertical) line from pole to pole (1) MP2: at least two further equidistant straight, (vertical) lines from pole to pole (1) MP3: arrow on any line, north to south (1)	ignore lines outside of the magnets for MP1 and MP2 judge by eye any arrow south to north, no mark awarded for MP3	(3)

Q10.

Question Number	Answer	Additional guidance	Mark
	 at least three straight lines between N and S poles (equally spaced) (1) **all** drawn arrows on lines are from N to S (1)	ignore drawn lines which start inside the shaded regions judge by eye lines do not have to touch magnets ignore lines drawn above or below magnets / edge effects for this mark	**(2)** **AO2.2**